

Mérida Urban Intelligence

Geospatial Data Warehouse · Technical report · Unit 2 Project

Business Intelligence · Data Engineering, 9th Quarter, Group B · Universidad Politécnica de Yucatán
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1 Problem and data sources

Mérida's demographic, economic, cartographic and public-safety data come from different institutions and describe the city in incompatible ways: census results are tables keyed by statistical areas, business and incident records are points, and boundaries are polygons. None of the sources can answer, on its own, whether population, economic activity and insecurity follow the same geography. The project integrates them into a single PostgreSQL/PostGIS warehouse that calculates territorial KPIs, compares areas and tests their spatial association; every number in this report is computed from that warehouse.

Layer	Source	Original grain	Records for Mérida
Demographic	INEGI Population and Housing Census 2020, results by urban AGEb and block	block, with AGEb totals	17,073 rows
Economic	INEGI DENUe (registrations up to 2026-04)	establishment (point)	56,909 establishments
Geographic	INEGI Marco Geoestadístico 2020	polygon	526 urban AGEbs
Public safety	SESNSP municipal crime incidence (no coordinates published for Mérida)	municipality × month × crime type	12,936 rows

2 Geographic integration strategy

The unit of analysis had to be one for which the census publishes results, so that points could be related to residents. Four candidates were measured on the downloaded data.

Candidate	Units	Median population	PEA withheld	Units without business	Decision
Block (manzana)	17,595	52	4.8%	33%	too small, unstable rates
Urban AGEb	526	1,755	0.8%	3%	selected
Colonia	—	—	—	—	no official polygons or census data
Locality	53	—	—	—	one locality holds 483 AGEbs

The study area is the 526 urban AGEbs of the municipality (259.9 km²), home to 957,399 of its 995,129 inhabitants (96.2%). Census and polygon keys match one to one. The cartography's projection has no EPSG code; its parameters were verified to equal EPSG:6372, where areas are measured, while geometries are stored in EPSG:4326 like the point sources. Points are assigned with a point-in-polygon rule (boundary points to the lowest key, unmatched points kept with a reason): **99.57%** of 56,909 establishments fall inside an AGEb, and **99.82%** of them agree with the AGEb that INEGI declares for the establishment, an independent confirmation of the join. Of the 245 unmatched, 165 are declared in rural areas.

3 Data Warehouse architecture

A batch pipeline moves the data **RAW** → **CLEAN** → **SPATIAL JOIN** → **PostgreSQL/PostGIS**. Raw files are checksummed and never edited; Python cleans and joins; SQL scripts rebuild a `staging` schema and the dimensional schema `dw`, create the KPI views and run 16 validation checks (15 pass, 1 warning) covering row counts, keys, referential integrity, reconciliation with the sources and geometry. A second run reproduces identical results.

The model is a **constellation schema**: the three AGEb-level fact tables share the conformed dimension `dim_geography`, which is what allows indicators that combine layers; the municipal crime counts, which have no location, form a fourth fact table on `dim_municipality` and are never assigned to AGEbs.

Fact	Grain
fact_demographics	one urban AGEb, census 2020

fact_establishment	one DENUE establishment
fact_crime_incident	one georeferenced incident
fact_crime_municipal	municipality × month × crime type

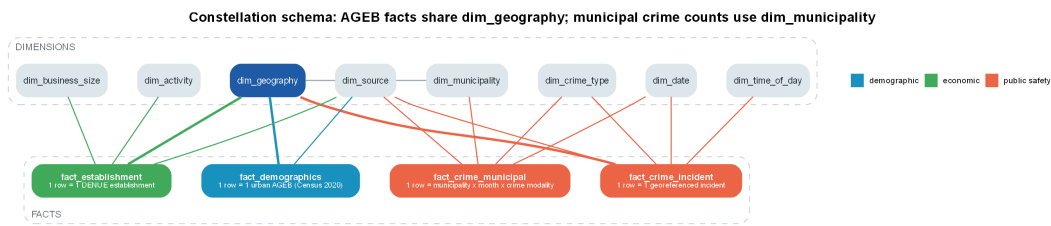


Figure 1. Facts and dimensions, coloured by thematic layer.

4 Key KPIs and spatial analysis

The fourteen required KPIs are SQL views with explicit rules: densities per km², rates in percent, and NULL instead of zero when a denominator is zero or unknown. AGEbS with fewer than 100 residents (32) are flagged because per-capita values there are extreme, and the 18 AGEbS where the census withholds a value keep it as missing.



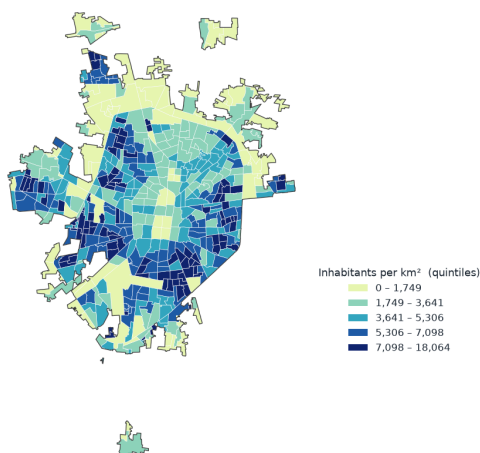
Retail trade is the dominant activity in 417 of the 526 AGEbS; 9.8% of residents are 65 or older. Relationships between indicators are measured with Spearman's ρ , robust to the skewed distributions, and checked with Pearson's r on log values. Spatial structure is measured with Moran's I under **Queen contiguity** (AGEbS sharing an edge or vertex are neighbours), row-standardised weights, log values and 999 permutations; the two detached AGEbS with no neighbour are excluded. Only touching AGEbS influence each other, so neighbours across a road reserve are ignored, and large peripheral AGEbS, with fewer neighbours, give less stable local results.

5 Main findings

Business activity concentrates in the centre; residents live around it. Population and business density are positively related ($\rho = 0.55$), but per resident the relation reverses: the densest residential AGEbS have fewer businesses per 1,000 inhabitants ($\rho = -0.31$).

Population density

Urban AGEbS of Mérida

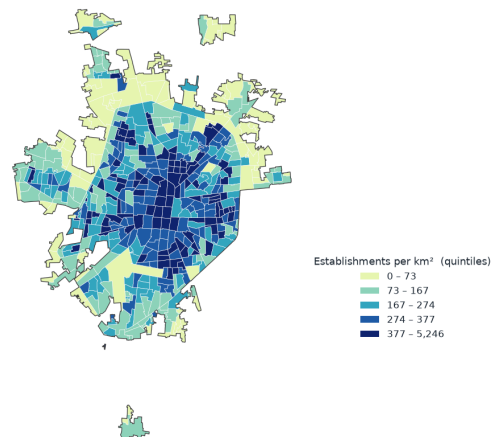


Source: INEGI Census 2020, DENUE and Marco Geoestadístico 2020 · computed in the warehouse

Figure 2. Population density (quintiles).

Business density

Urban AGEbS of Mérida



Source: INEGI Census 2020, DENUE and Marco Geoestadístico 2020 · computed in the warehouse

Figure 3. Business density (quintiles).

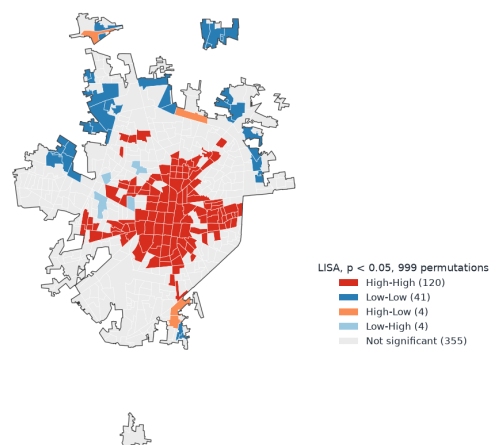
Indicators cluster strongly in space. All tested indicators show positive and significant global Moran's I; business density forms 120 High-High AGEBs over the historic centre and its extensions and 41 Low-Low AGEBs on the northern periphery.

Indicator	n	Moran's I	z	p
Population density	524	0.368	12.6	0.001
Business density	524	0.505	17.1	0.001
Economically active population rate	491	0.547	18.1	0.001
Businesses per 1,000 residents	491	0.455	15.8	0.001

x	y	n	ρ	r (log)	p
Population density	Business density	526	+0.55	+0.58	< 0.001
Population density	Businesses per 1,000 residents	494	-0.31	-0.19	< 0.001
Economically active population rate	Businesses per 1,000 residents	494	-0.35	-0.31	< 0.001
Population aged 65+	Service density	491	+0.57	+0.43	< 0.001

Local clusters of business density

Local Moran's I (LISA), Queen contiguity, log values



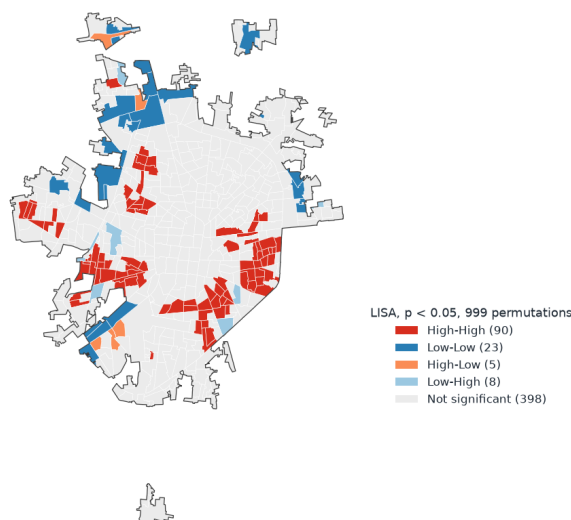
Source: INEGI Census 2020, DENUE and Marco Geoestadístico 2020 - computed in the warehouse

Figure 4. LISA clusters of business density ($p < 0.05$).

Business activity sits next to populated areas, and services next to older ones. Bivariate Moran's I between business density and the population density of neighbouring AGEBs is 0.32 ($p = 0.001$); AGEBs with older populations have higher service density ($\rho = 0.57$), and AGEBs with a higher economically active rate have fewer businesses per resident ($\rho = -0.35$), the profile of newer dormitory neighbourhoods.

Business density vs neighbouring population density

Bivariate LISA: value of x in the AGEB against the mean of y in its neighbours



Source: INEGI Census 2020, DENUE and Marco Geoestadístico 2020 - computed in the warehouse

Figure 5. Bivariate LISA: business density against neighbouring population density.

Public safety at municipal level

No dataset with incident coordinates is published for Mérida; the most detailed official source is the monthly municipal incidence of the SESNSP. It enters the warehouse as a separate fact table at municipality $\times$ month $\times$ crime-type grain. In 2025 the municipality recorded **3,135** incidents of the common jurisdiction, **3.1** per 1,000 residents and **5.5** per 100

establishments; 41% were property crimes. Because these counts have no location, they cannot be assigned to AGEBS, and the AGEB-level crime KPIs and spatial statistics remain empty by design; the point-level path is implemented and tested for when georeferenced data become available.

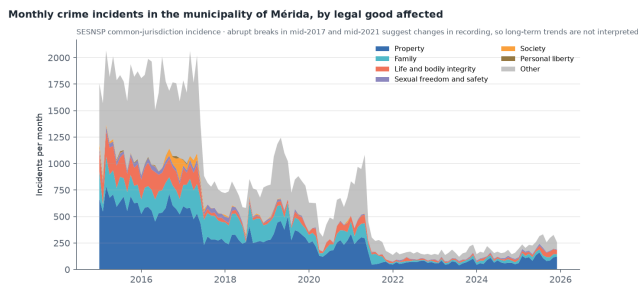


Figure 6. Monthly incidents by legal good affected, 2015–2025 (SESNP).

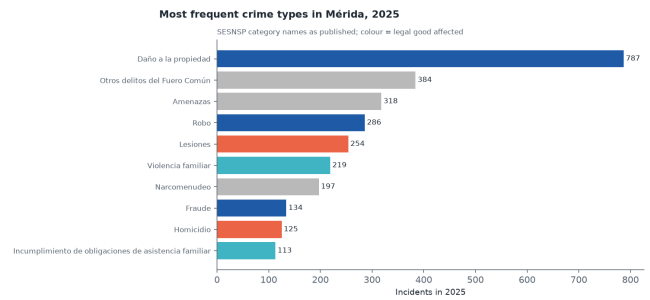


Figure 7. Most frequent crime types in 2025.

6 Limitations and interpretation cautions

- **Association, not causation.** Correlations and Moran statistics describe how areas co-vary in space.
- **Spatial dependence.** Because neighbouring AGEBS are similar, ordinary p-values of the correlations overstate the evidence.
- **Ecological fallacy and MAUP.** Results describe AGEBS, not people or firms, and could change with other boundaries.
- **Different reference dates.** Population and boundaries are from 2020, establishments from the latest DENU, crime from 2025.
- **Coverage.** DENU under-represents informal and home-based activity; 245 establishments outside the 2020 urban AGEBS are excluded.
- **Public safety.** Without published incident coordinates, crime cannot be analysed inside the city; the warehouse is ready for a point-level source. The SESNP series shows abrupt breaks in mid-2017 and mid-2021 that suggest changes in recording, so trends are not interpreted.
- **Neighbourhood rule.** Queen contiguity ignores proximity across gaps; another rule could change which AGEBS are flagged locally.